

- 4.11** The process of assigning load addresses to the various parts of the program and adjusting the code and data in the program to reflect the assigned addresses is called
- (a) assembly
 - (b) parsing
 - (c) relocation
 - (d) symbol resolution

[2001 : 1 M]

- 4.12** Consider a virtual memory system with FIFO page replacement policy. For an arbitrary page access pattern, increasing the number of page frames in main memory will
- (a) always decrease the number of page faults
 - (b) always increase the number of page faults
 - (c) sometimes increase the number of page faults
 - (d) never affect the number of page faults

[2001 : 1 M]

- 4.13** Which of the following is not a form of memory?
- (a) Instruction cache
 - (b) Instruction register
 - (c) Instruction opcode
 - (d) Translation look a side buffer

[2002 : 1 M]

- 4.14** The optimal page replacement algorithm will, select the page that
- (a) Has not been used for the longest time in the past
 - (b) Will not be used for the longest time in the future
 - (c) Has been used least number of times
 - (d) Has been used most number of times

[2002 : 1 M]

- 4.15** Dynamic linking can cause security concerns because
- (a) Security is dynamic
 - (b) The path for searching dynamic libraries is not known till runtime
 - (c) Linking is insecure
 - (d) Cryptographic procedures are not available for dynamic linking

[2002 : 2 M]

- 4.16** In a system with 32 bit virtual addresses and 1 KB page size, use of one-level page tables for virtual to physical address translation is not practical because of
- (a) the large amount of internal fragmentation
 - (b) the large amount of external fragmentation
 - (c) the large memory overhead in maintaining page tables
 - (d) the large computation overhead in the translation process

[2003 : 1 M]

Common Data for Q. 4.17 & 4.18:

A processor uses 2-level page tables for virtual to physical address translation. Page tables for both levels are stored in the main memory. Virtual and physical addresses are both 32 bits wide. The memory is byte addressable. For virtual to physical address translation, the 10 most significant bits of the virtual address are used as index into the first level page table while the next 10 bits are used as index into the second level page table. The 12 least significant bits of the virtual address are used as offset within the page. Assume that the page table entries in both levels of page tables are 4 bytes wide. Further, the processor has a translation look-aside buffer (TLB), with a hit rate of 96%. The TLB caches recently used virtual page numbers and the corresponding physical page numbers. The processor also has a physically addressed cache with a hit ratio of 90%. Main memory access time is 10 ns, cache access time is 1 ns, and TLB access time is also 1 ns.

- 4.17** Assuming that no page faults occur, the average time taken to access a virtual address is approximately (to the nearest 0.5 ns)
- (a) 1.5 ns
 - (b) 2 ns
 - (c) 3 ns
 - (d) 4 ns

[2003 : 2 M]

- 4.18** Suppose a process has only the following pages in its virtual address space: two contiguous code pages starting at virtual address 0×00000000 , two contiguous data pages starting at virtual address 0×00400000 , and a stack page starting at virtual address $0 \times FFFFF000$. The amount of memory required for storing the page tables of this process is

- (a) 8 KB
- (b) 12 KB
- (c) 16 KB
- (d) 20 KB

[2003 : 2 M]

- 4.19** Consider an operating system capable of loading and executing a single sequential user process at a time. The disk head scheduling algorithm used is First Come First Served (FCFS). If FCFS is replaced by Shortest Seek Time First (SSTF), claimed by the vendor to given 50% better benchmark results, what is the expected improvement in the I/O performance of user programs?

- (a) 50%
- (b) 40%
- (c) 25%
- (d) 0%

[2004 : 1 M]

- 4.20** The minimum number of page frames that must be allocated to a running process in a virtual memory environment is determined by

- (a) the instruction set architecture
- (b) page size
- (c) physical memory size
- (d) number of processes in memory

[2004 : 1 M]

4.21 Consider a system with a two-level paging scheme in which a regular memory access takes 150 nanoseconds, and servicing a page fault takes 8 milliseconds. An average instruction takes 100 nanoseconds of CPU time, and two memory accesses. The TLB hit ratio is 90%, and the page fault rate is one in every 10,000 instructions. What is the effective average instruction execution time?

- (a) 645 nanoseconds
- (b) 1050 nanoseconds
- (c) 1215 nanoseconds
- (d) 1260 nanoseconds

[2004 : 2 M]

4.22 Consider a fully associative cache with 8 cache blocks (numbered 0-7) and the following sequence of memory block requests: 4, 3, 25, 8, 19, 6, 25, 8, 16, 35, 45, 22, 8, 3, 16, 25, 7.

If LRU replacement policy is used, which cache block will have memory block 7?

- (a) 4
- (b) 5
- (c) 6
- (d) 7

[2004 : 2 M]

4.23 In a virtual memory system, size of virtual address is 32-bit, size of physical address is 30-bit, page size is 4 Kbyte and size of each page table entry is 32-bit. The main memory is byte addressable. Which one of the following is the maximum number of bits that can be used for storing protection and other information in each page table entry?

- (a) 2
- (b) 10
- (c) 12
- (d) 14

[2004 : 2 M]

4.24 For each of the four processes P_1 , P_2 , P_3 and P_4 . The total size in kilobytes (KB) and the number of segments are given below.

Process	Total size (in KB)	Number of segments
P_1	195	4
P_2	254	5
P_3	45	3
P_4	364	8

The page size is 1 KB. The size of an entry in the page table is 4 bytes. The size of an entry in the segment table is 8 bytes. The maximum size of a segment is 256 KB. The paging method for memory management uses two-level paging, and

its storage overhead is P. The storage overhead for the segmentation method is S. The storage overhead for the segmentation and paging method is T. What is the relation among the overheads for the different methods of memory management in the concurrent execution of the above four processes?

- (a) $P < S < T$
- (b) $S < P < T$
- (c) $S < T < P$
- (d) $T < S < P$

[2006 : 2 M]

4.25 A CPU generates 32-bit virtual addresses. The page size is 4 KB. The processor has a translation look-aside buffer (TLB) which can hold a total of 128 page table entries and is 4-way set associative. The minimum size of the TLB tag is

- (a) 11 bits
- (b) 13 bits
- (c) 15 bits
- (d) 20 bits

[2006 : 2 M]

4.26 A computer system supports 32-bit virtual addresses as well as 32-bit physical addresses. Since the virtual address space is of the same size as the physical address space, the operating system designers decide to get rid of the virtual memory entirely. Which one of the following is true?

- (a) Efficient implementation of multi-user support is no longer possible
- (b) The processor cache organization can be made more efficient now
- (c) Hardware support for memory management is no longer needed
- (d) CPU scheduling can be made more efficient now

[2006 : 2 M]

4.27 Let a memory have four free blocks of sizes 4k, 8k, 20k, 2k. These blocks are allocated following the best-fit strategy. The allocation requests are stored in a queue as shown below:

Request No.	Request sizes	Usage Time
J1	2k	4
J2	14k	10
J3	3k	2
J4	6k	8
J5	6k	4
J6	10k	1
J7	7k	8
J8	20k	6

The time at which the request for J7 will be completed will be

- (a) 16
- (b) 19
- (c) 20
- (d) 37

[2007 : 1 M]

4.28 The address sequence generated by tracing a particular program executing in a pure demand paging system with 100 bytes per page is 0100, 0200, 0430, 0499, 0510, 0530, 0560, 0120, 0220, 0240, 0260, 0320, 0410.

Suppose that the memory can store only one page and if x is the address which causes a page fault then the bytes from addresses x to $x + 99$ are loaded on to the memory. How many page faults will occur?

- (a) 0 (b) 4
(c) 7 (d) 8

[2007 : 1 M]

4.29 A demand paging system takes 100 time units to service a page fault and 300 time units to replace a dirty page. Memory access time is 1 time unit. The probability of a page fault is p . In case of a page fault, the probability of page being dirty is also p . It is observed that the average access time is 3 time units. Then the value of p is

- (a) 0.194 (b) 0.233
(c) 0.514 (d) 0.981

[2007 : 2 M]

Common Data for Q. 4.30 & Q. 4.31:

A process has been allocated 3 page frames. Assume that none of the pages of the process are available in the memory initially. The process makes the following sequence of page references (reference string) : 1, 2, 1, 3, 7, 4, 5, 6, 3, 1.

4.30 If optimal page replacement policy is used, how many page faults occur for the above reference string?

- (a) 7 (b) 8
(c) 9 (d) 10

[2007 : 2 M]

4.31 Least Recently Used (LRU) page replacement policy is a practical approximation to optimal page replacement. For the above reference string, how many more page faults occur with LRU than with the optimal page replacement policy?

- (a) 0 (b) 1
(c) 2 (d) 3

[2007 : 2 M]

4.32 A paging scheme uses a Translation Look-aside Buffer (TLB). A TLB-access takes 10 ns and a main memory access takes 50 ns. What is the effective access time (in ns) if the TLB hit ratio is 90% and there is no page-fault?

- (a) 54 (b) 60
(c) 65 (d) 75

[2008 : 1 M]

4.33 A processor uses 36 bit physical addresses and 32 bit virtual addresses, with a page frame size of 4 Kbytes. Each page table entry is of size 4 bytes. A three level page table is used for virtual-to-physical address translation, where the virtual address is used as follows:

- bits 30-31 are used to index into the first level page table,
- bits 21-29 are used to index into the second level page table
- bits 12-20 are used to index into the third level page table
- bits 0-11 are used as offset within the page

The number of bits required for addressing the next level page table (or page frame) in the page table entry of the first, second and third level page tables are respectively.

- (a) 20, 20 and 20 (b) 24, 24, and 24
(c) 24, 24 and 20 (d) 25, 25 and 24

[2008 : 2 M]

4.34 Assume that a main memory with only 4 pages, each of 16 bytes, is initially empty. The CPU generates the following sequence of virtual addresses and uses the Least Recently Used (LRU) page replacement policy.

0, 4, 8, 20, 24, 36, 44, 12, 68,
72, 80, 84, 28, 32, 88, 92

How many page faults does this sequence cause? What are the page numbers of the pages present in the main memory at the end of the sequence?

- (a) 6 and 1, 2, 3, 4 (b) 7 and 1, 2, 4, 5
(c) 8 and 1, 2, 4, 5 (d) 9 and 1, 2, 3, 5

[2008 : 2 M]

4.35 Match the following flag bits used in the context of virtual memory management on the List-I (Name of the bit) with the different purposes on the List-II (Purpose) of the table below:

List-I	List-II
I. Dirty	a. Page initialization
II. R/W	b. Write-back policy
III. Reference	c. Page protection
IV. Valid	d. Page replacement policy

Codes:

- (a) I-d, II-a, III-b, IV-c (b) I-b, II-c, III-a, IV-d
(c) I-c, II-d, III-a, IV-b (d) I-b, II-c, III-d, IV-a

[2008 : 2 M]

4.36 In which one of the following page replacement policies, Bledy's anomaly may occur?

- (a) FIFO (b) Optimal
(c) LRU (d) MRU

[2009 : 1 M]

- 4.37** The essential content(s) in each entry of a page table is/are
 (a) virtual page number
 (b) page frame number
 (c) both virtual page number and page frame number
 (d) access right information [2009 : 1 M]

- 4.38** A system uses FIFO policy for page replacement. It has 4 page frames with no pages loaded to begin with. The system first accesses 100 distinct pages in some order and then accesses the same 100 pages but now in the reverse order. How many page faults will occur?
 (a) 196 (b) 192
 (c) 197 (d) 195 [2010 : 1 M]

- 4.39** Let the page fault service time be 10 ms in a computer with average memory access time being 20 ns. If one page fault is generated for every 10^6 memory accesses, what is the effective access time for the memory?
 (a) 21 ns (b) 30 ns
 (c) 23 ns (d) 35 ns [2011 : 1 M]

- 4.40** Consider the virtual page reference string
 1, 2, 3, 2, 4, 1, 3, 2, 4, 1
 on a demand paged virtual memory system running on computer system that has main memory size of 3 page frames which are initially empty. Let LRU, FIFO and OPTIMAL denote the number of page faults under the corresponding page replacement policy. Then
 (a) OPTIMAL < LRU < FIFO
 (b) OPTIMAL < FIFO < LRU
 (c) OPTIMAL = LRU
 (d) OPTIMAL = FIFO [2012 : 2 M]

Linked Answer Questions 4.41 and 4.42:

A computer uses 46-bit virtual address, 32-bit physical address, and a three-level paged page table organization. The page table base register stores the base address of the first-level (T_1), which occupies exactly one page. Each entry of T_1 stores the base address of a page of the second-level table (T_2). Each entry of T_2 stores the base address of a page of the third-level table (T_3). Each entry of T_3 stores a page table entry (PTE). The PTE is 32 bit in size. The processor used in the computer has a 1 MB 16-way-set associative virtually indexed physical tagged cache the cache block size is 64 bytes.

- 4.41** What is the size of a page in KB in this computer?
 (a) 2 (b) 4
 (c) 8 (d) 16 [2013 : 2 M]

- 4.42** What is the minimum number of page colours needed to guarantee that no two synonyms map to different sets in the processor cache of this computer?
 (a) 2 (b) 4
 (c) 8 (d) 16 [2013 : 2 M]

- 4.43** Assume that there are 3 page frames which are initially empty. If the page reference string is 1, 2, 3, 4, 2, 1, 5, 3, 2, 4, 6, the number of page faults using the *optimal replacement policy* is _____. [2014 (Set-1) : 2 M]

- 4.44** A computer has twenty physical page frames which contain pages numbered 101 through 120. Now a program accesses the pages numbered 1, 2, ..., 100 in that order, and repeats the access sequence THRICE.

Which one of the following page replacement policies experiences the same number of page faults as the optimal page replacement policy for this program?

- (a) Least-recently-used (b) First-in-first-out
 (c) Last-in-first-out (d) Most-recently-used [2014 (Set-2) : 2 M]

- 4.45** A system uses 3 page frames for storing process pages in main memory. It uses the Least Recently Used (LRU) page replacement policy. Assume that all the page frames are initially empty. What is the total number of page faults that will occur while processing the page reference string given below?

4, 7, 6, 1, 7, 6, 1, 2, 7, 2

[2014 (Set-3) : 1 M]

- 4.46** Consider a paging hardware with a TLB. Assume that the entire page table and all the pages are in the physical memory. It takes 10 milliseconds to search the TLB and 80 milliseconds to access the physical memory. If the TLB hit ratio is 0.6, the effective memory access time (in milliseconds) is _____. [2014 (Set-3) : 2 M]

- 4.47** Consider a system with byte-addressable memory, 32 bit logical addresses, 4 kilobyte page size and page table entries of 4 bytes each. The size of the page table in the system in megabytes is _____. [2015 (Set-1) : 1 M]

- 4.48** Consider a main memory with five page frames and the following sequence of page references:

3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3

Which one of the following is true with respect to page replacement policies First-In-First Out (FIFO) and Least Recently Used (LRU)?

- (a) Both incur the same number of page faults
- (b) FIFO incurs 2 more page faults than LRU
- (c) LRU incurs 2 more page faults than FIFO
- (d) FIFO incurs 1 more page faults than LRU

[2015 (Set-1) : 2 M]

4.49 A computer system implements a 40 bit virtual address, page size of 8 kilobytes, and a 128-entry translation look-aside buffer (TLB) organized into 32 sets each having four ways. Assume that the TLB tag does not store any process id. The minimum length of the TLB tag in bits is ____.

[2015 (Set-2) : 1 M]

4.50 Consider six memory partitions of size 200 KB, 400 KB, 600 KB, 500 KB, 300 KB, and 250 KB, where KB refers to kilobyte. These partitions need to be allotted to four processes of sizes 357 KB, 210 KB, 468 KB and 491 KB in that order. If the best fit algorithm is used, which partitions are NOT allotted to any process?

- (a) 200 KB and 300 KB
- (b) 200 KB and 250 KB
- (c) 250 KB and 300 KB
- (d) 300 KB and 400 KB

[2015 (Set-2) : 2 M]

4.51 A Computer system implements 8 kilobyte pages and a 32-bit physical address space. Each page table entry contains a valid bit, a dirty bit, three permission bits, and the translation. If the maximum size of the page table of a process is 24 megabytes, the length of the virtual address supported by the system is ____ bits.

[2015 (Set-2) : 2 M]

4.52 Consider a computer system with 40-bit virtual addressing and page size of sixteen kilobytes. If the computer system has a one-level page table per process and each page table entry requires 48 bits, then the size of the per-process page table is ____ megabytes.

[2016 (Set-1) : 2 M]

4.53 Consider a computer system with ten physical page frames. The system is provided with an access sequence $(a_1, a_2, \dots, a_{20}, a_1, a_2, \dots, a_{20})$, where each a_i is a distinct virtual page number. The difference in the number of page faults between the last-in-first-out page replacement policy and the optimal page replacement policy is ____.

[2016 (Set-1) : 2 M]

4.54 In which one of the following page replacement algorithms it is possible for the page fault rate to increase even when the number of allocated frames increases?

- (a) LRU (Least Recently Used)
- (b) OPT (Optimal Page Replacement)
- (c) MRU (Most Recently Used)
- (d) FIFO (First In First Out)

[2016 (Set-2) : 1 M]

4.55 Recall that Belady's anomaly is that the page-fault rate may increase as the number of allocated frames increases. Now, consider the following statements:

S1: Random page replacement algorithm (where a page chosen at random is replaced) suffers from Belady's anomaly

S2: LRU page replacement algorithm suffers from Belady's anomaly

Which of the following is CORRECT?

- (a) S1 is true, S2 is true
- (b) S1 is true, S2 is false
- (c) S1 is false, S2 is true
- (d) S1 is false, S2 is false

[2017 (Set-1) : 2 M]

4.56 Consider a process executing on an operating system that uses demand paging. The average time for a memory access in the system is M units if the corresponding memory page is available in memory and D units if the memory access causes a page fault. It has been experimentally measured that the average time taken for a memory access in the process is X units.

Which one of the following is the correct expression for the page fault rate experienced by the process?

- | | |
|-------------------------------|-------------------------------|
| (a) $\frac{(D - M)}{(X - M)}$ | (b) $\frac{(X - M)}{(D - M)}$ |
| (c) $\frac{(D - X)}{(D - M)}$ | (d) $\frac{(X - M)}{(D - X)}$ |

[2018 : 1 M]

4.57 Assume that in a certain computer, the virtual addresses are 64 bits long and the physical addresses are 48 bits long. The memory is word addressable. The page size is 8 kB and the word size is 4 bytes. The Translation Look-aside Buffer (TLB) in the address translation path has 128 valid entries. At most how many distinct virtual addresses can be translated without any TLB miss?

- (a) 16×2^{10}
- (b) 8×2^{20}
- (c) 4×2^{20}
- (d) 256×2^{10}

[2019 : 2 M]

4.58 Consider allocation of memory to a new process. Assume that none of the existing holes in the memory will exactly fit the process's memory requirement. Hence, a new hole of smaller size will be created if allocation is made in any of the existing holes. Which one of the following statements is TRUE?

- (a) The hole created by next fit is never larger than the hole created by best fit.
- (b) The hole created by first fit is always larger than the hole created by next fit.
- (c) The hole created by best fit is never larger than the hole created by first fit.
- (d) The hole created by worst fit is always larger than the hole created by first fit.

[2020 : 1 M]

4.59 Consider a paging system that uses 1-level page table residing in main memory and a TLB for address translation. Each main memory access takes 100 ns and TLB lookup takes 20 ns. Each page transfer to/from the disk takes 5000 ns. Assume that the TLB hit ratio is 95%, page fault rate is 10%. Assume that for 20% of the total page faults, a dirty page has to be written back to disk before the required page is read in from disk. TLB update time is negligible. The average memory access time in ns (round off to 1 decimal places) is _____.

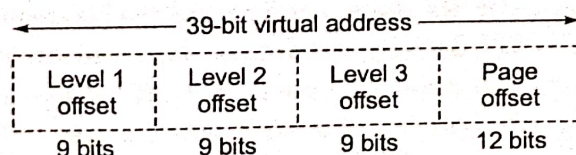
[2020 : 2 M]

4.60 In the context of operating systems, which of the following statements is/are correct with respect to paging?

- (a) Paging incurs memory overheads.
- (b) Multi-level paging is necessary to support pages of different sizes.
- (c) Page size has no impact on internal fragmentation.
- (d) Paging helps solve the issue of external fragmentation.

[2021 (Set-1) : 1 M]

4.61 Consider a three-level page table to translate a 39-bit virtual address to a physical address as shown below:



The page size is 4 KB ($1 \text{ KB} = 2^{10} \text{ bytes}$) and page table entry size at every level is 8 bytes. A process P is currently using 2 GB ($1 \text{ GB} = 2^{30} \text{ bytes}$)

virtual memory which is mapped to 2 GB of physical memory. The minimum amount of memory required for the page table of P across all levels is _____ KB.

[2021 (Set-2) : 2 M]

4.62 Which one of the following statements is FALSE?

- (a) The TLB performs an associative search in parallel on all its valid entries using page number of incoming virtual address.
- (b) If the virtual address of a word given by CPU has a TLB hit, but the subsequent search for the word results in a cache miss, then the word will always be present in the main memory.
- (c) The memory access time using a given inverted page table is always same for all incoming virtual addresses.
- (d) In a system that uses hashed page tables, if two distinct virtual addresses V_1 and V_2 map to the same value while hashing, then the memory access time of these addresses will not be the same.

[2022 : 2 M]

4.63 Consider a demand paging system with four page frames (initially empty) and LRU page replacement policy. For the following page reference string

7, 2, 7, 3, 2, 5, 3, 4, 6, 7, 7, 1, 5, 6, 1

the page fault rate, defined as the ratio of number of page faults to the number of memory accesses (rounded off to one decimal place) is _____.

[2022 : 2 M]

4.64 Consider the following two-dimensional array D in the C programming language, which is stored in row-major order:

```
int D[128][128];
```

Demand paging is used for allocating memory and each physical page frame holds 512 elements of the array D . The Least Recently Used (LRU) page-replacement policy is used by the operating system. A total of 30 physical page frames are allocated to a process which executes the following code snippet:

```
for (int i = 0; i < 128; i++)
    for (int j = 0; j < 128; j++)
        D[j][i] *= 10;
```

The number of page faults generated during the execution of this code snippet is _____.

[2023 : 2 M]

4.65 Consider a computer system with 57-bit virtual addressing using multi-level tree-structured page tables with L levels for virtual to physical address translation. The page size is 4 KB (1 KB = 1024 B) and a page table entry at any of the levels occupies 8 bytes. The value of L is _____. [2023 : 2 M]

4.66 Consider a memory management system that uses a page size of 2 KB. Assume that both the physical and virtual addresses start from 0. Assume that the pages 0, 1, 2, and 3 are stored in the page frames 1, 3, 2, and 0, respectively. The physical address (in decimal format) corresponding to the virtual address 2500 (in decimal format) is _____. [2024 (Set-1) : 2 M]

4.67 Which of the following tasks is/are the responsibility/responsibilities of the memory management unit (MMU) in a system with paging-based memory management?

- (a) Allocate a new page table for a newly created process.
- (b) Translate a virtual address to a physical address using the page table.
- (c) Raise a trap when a process tries to write to a page marked with read-only permission in the page table.
- (d) Raise a trap when a virtual address is not found in the page table.

[2024 (Set-2) : 1 M]

4.68 Consider a 32-bit system with 4 KB page size and page table entries of size 4 bytes each. Assume 1 KB = 2^{10} bytes. The OS uses a 2-level page table for memory management, with the page table containing an outer page directory and an inner page table. The OS allocates a page for the outer page directory upon process creation. The OS uses demand paging when allocating memory for the inner page table, i.e., a page of the inner page table is allocated only if it contains at least one valid page table entry.

An active process in this system accesses 2000 unique pages during its execution, and none of the pages are swapped out to disk. After it completes the page accesses, let X denote the minimum and Y denote the maximum number of pages across the two levels of the page table of the process.

The value of $X + Y$ is _____. [2024 (Set-2) : 2 M]

4.69 Consider a demand paging memory management system with 32-bit logical address, 20-bit physical address, and page size of 2048 bytes. Assuming that the memory is byte addressable, what is the maximum number of entries in the page table?

- (a) 2^{21}
- (b) 2^{20}
- (c) 2^{22}
- (d) 2^{24}

[2025 (Set-1) : 1 M]

4.70 In optimal page replacement algorithm, information about all future page references is available to the operating system (OS). A modification of the optimal page replacement algorithm is as follows:

The OS correctly predicts only up to next 4 page references (including the current page) at the time of allocating a frame to a page.

A process accesses the pages in the following order of page numbers:

1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4

If the system has three memory frames that are initially empty, the number of page faults that will occur during execution of the process is _____. (Answer in integer)

[2025 (Set-1) : 2 M]

4.71 Consider a demand paging system with three frames, and the following page reference string: 1 2 3 4 5 4 1 6 4 5 1 3 2. The contents of the frames are as follows initially and after each reference (from left to right):

Initially	after											
-	1*	2*	4*	5*	4	1	6*	4	5	1*	3*	2*
-	1	1	1	1	1	1	6	6	6	6	6	2
-	-	2	2	4	4	4	4	4	4	1	1	1
-	-	-	3	3	5	5	5	5	5	5	3	3

The *-marked references cause page replacements.

Which one or more of the following could be the page replacement policy/policies in use?

- (a) Least Recently Used page replacement policy
- (b) Least Frequently Used page replacement policy
- (c) Most Frequently Used page replacement policy
- (d) Optimal page replacement policy

[2025 (Set-2) : 2 M]

4.72 A computer system supports a logical address space of 2^{32} bytes. It uses two-level hierarchical paging with a page size of 4096 bytes. A logical address is divided into a b -bit index to the outer page table, an offset within the page of the inner page table, and an offset within the desired page. Each entry of the inner page table uses eight bytes. All the pages in the system have the same size.

The value of b is _____. (Answer in integer)

[2025 (Set-2) : 2 M]

4.73 Consider a system that has a cache memory unit and a memory management unit (MMU). The address input to the cache memory is a physical address. The MMU has a translation lookaside buffer (TLB). Assume that when a page is evicted from the main memory, the corresponding blocks in the cache are marked as invalid.

For a given memory reference, which of the following sequences of events can NEVER happen?

- (a) TLB miss, Page table hit, Cache hit
- (b) TLB hit, Page table miss, Cache hit
- (c) TLB miss, Page table miss, Cache hit
- (d) TLB miss, Page table miss, Cache miss

[2026 (Set-1) : 2 M]

4.74 A system has a Translation Lookaside Buffer (TLB) that has a reach of 1 MB. TLB reach is defined as the total amount of physical memory that can be accessed through the TLB entries. The paging system uses pages of size 4 KB. The virtual address space is 64 GB and physical address space is 1 GB. If each TLB entry stores a 4-bit process id, page number, frame number, and a 2-bit control field, then the size of the TLB (in bytes) is _____. (answer in integer)

Note: $1K = 2^{10}$, $1M = 2^{20}$, $1G = 2^{30}$

[2026 (Set-2) : 2 M]

4.75 Consider contiguous allocation of physical memory to processes using variable partitioning scheme. Suppose there are 8 holes in the memory of sizes 20 KB, 4 KB, 25 KB, 18 KB, 7 KB, 9 KB, 15 KB, and 12 KB. Assume that no two holes are adjacent. Two processes P1 of size 16 KB and P2 of size 9 KB arrive in that order, and they are allocated memory using the best-fit technique. After allocating space to P1 and P2, the number of holes of size less than 8 KB is _____. (answer in integer)

Note: $1K = 2^{10}$

[2026 (Set-2) : 2 M]

■■■■

Answers Memory Management and Virtual Memory

4.1 (b)	4.2 (c)	4.3 (a)	4.4 (c)	4.5 (b)	4.6 (a)	4.7 (d)	4.8 (d)	4.9 (c)
4.10 (d)	4.11 (c)	4.12 (c)	4.13 (c)	4.14 (b)	4.15 (b)	4.16 (c)	4.17 (d)	4.18 (c)
4.19 (d)	4.20 (a)	4.21 (d)	4.22 (b)	4.23 (d)	4.24 (b)	4.25 (c)	4.26 (a)	4.27 (b)
4.28 (c)	4.29 (a)	4.30 (a)	4.31 (c)	4.32 (c)	4.33 (b)	4.34 (b)	4.35 (d)	4.36 (a)
4.37 (b)	4.38 (a)	4.39 (b)	4.40 (b)	4.41 (c)	4.42 (c)	4.43 (7)	4.44 (d)	4.45 (6)
4.46 (122)	4.47 (4)	4.48 (a)	4.49 (22)	4.50 (a)	4.51 (36)	4.52 (384)	4.53 (1)	4.54 (d)
4.55 (b)	4.56 (b)	4.57 (d)	4.58 (c)	4.59 (154.5)	4.60 (a, d)	4.61 (4108)	4.62 (c)	4.63 (0.6)
4.64 (4096)	4.65 (5)	4.66 (6596)	4.67 (b, c, d)	4.68 (1028)	4.69 (a)	4.70 (6)	4.71 (d)	4.72 (11)
4.73 (b, c)	4.74 (1536)	4.75 (3)						

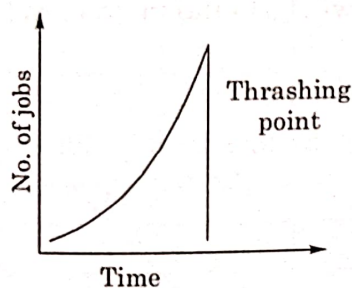
Explanations Memory Management and Virtual Memory

4.1 (b)

Locality of reference is a phenomenon describing the same value, related storage location being frequently accessed.

4.2 (c)

Thrashing: After reaching the maximum number of jobs allowed in main memory. CPU efficiency drastically getting down this situation is called Thrashing.



Thrashing implies excessive page I/O.

4.3 (a)

Which page has Dirty bit only those pages being written back in main memory, so it avoids unnecessary writing.

4.4 (c)

Loader is used to load the program into main memory.

4.5 (b)

Maximum space that can be needed at a time is 14 kB (using E). The maximum space to load the program is decided by the maximum size taken by routine call sequence from the root. i.e., Root $\rightarrow$ A $\rightarrow$ E.

4.6 (a)

$$\text{EMAT} = p \times s + (1 - p) \times m$$

p: page fault rate

s: page fault service time

m: main memory access time

$$\Rightarrow \frac{(i+j)}{k} + \left(1 - \frac{1}{k}\right) \times i = i + \frac{j}{k}$$

4.7 (d)

Virtual memory allows the user can run the programs larger than the physical memory size.

4.8 (d)

Average memory access time = [(% of page miss) * (time of service a page fault) + (% of page hit) * (memory access time)]/100
So, average memory access time in microseconds is

$$\begin{aligned} & \frac{(99.99 * 1 + 0.01 * 10 * 1000)}{100} \\ &= \frac{(99.99 + 100)}{1000} \\ &= \frac{199.99}{1000} = 1.9999 \mu\text{sec} \end{aligned}$$

4.9 (c)

$$\text{Number of entries in page table} = \frac{2^{32}}{2^{12}} = 2^{20}$$

$$\text{Frame size} = \frac{2^{26}}{2^{12}} = 2^{14}$$

$\therefore$ PT have to be stored in one frame so entry size must be 2 bytes, hence size of PT
 $= 2^{20} \times 2 = 2^{21} = 2 \text{ MB}$

4.10 (d)

In a system with virtual memory context switch includes extra overhead in switching of address spaces.

4.11 (c)

Relocation can change the assigned address of data and code in the program.

4.12 (c)

Bleady's Anomaly increasing number of frames = increase in page faults.

4.13 (c)

Instruction cache, instruction register and TLB are memories, but instruction opcode is the part of instruction that specify the operation to perform.

4.14 (b)

Optimal page replacement algorithm select a page that will not be used in the longest time in future.

4.15 (b)

In dynamic linking path for searching dynamic libraries is not known till runtime i.e. keep on changing, which can create security concern.

4.16 (c)

Given, Page Size = 1 KB = 2^{10} bytes

Virtual address is 32 bit long

So required number of pages

$$= \frac{2^{32}}{2^{10}} = 2^{22}$$

2^{22} is a very large so the large memory overhead in maintaining page tables.

4.17 (d)

Average time taken to access a virtual address

$$\begin{aligned} &= \left[\left(\frac{96}{100} \right) (1 + 0.9 \times 1 + 0.1 \times (1 + 10)) \right] \\ &\quad + [(0.04)(21 + 0.9 \times 1 + 0.1 \times (1 + 10))] \\ &= [(.96)(1 + 0.9 + 0.1 \times 11)] \\ &\quad + [(0.04)(21 + 0.9 + 0.1 \times 11)] \\ &= 3.87 \text{ ns} = 4 \text{ ns} \end{aligned}$$

4.18 (c)

Logical address = 32 bits

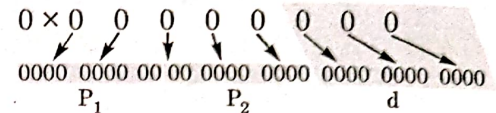
P_1	P_2	d
10 bits	10 bits	12 bits

P_1 = 1st level page table

P_2 = 2nd level page table

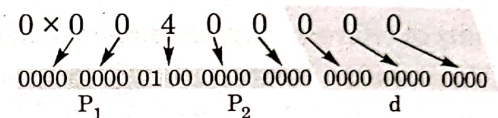
d = Page offset

First virtual address given



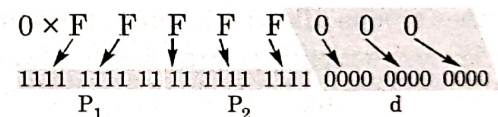
Address will be found in first level page table first block.

Second virtual address given

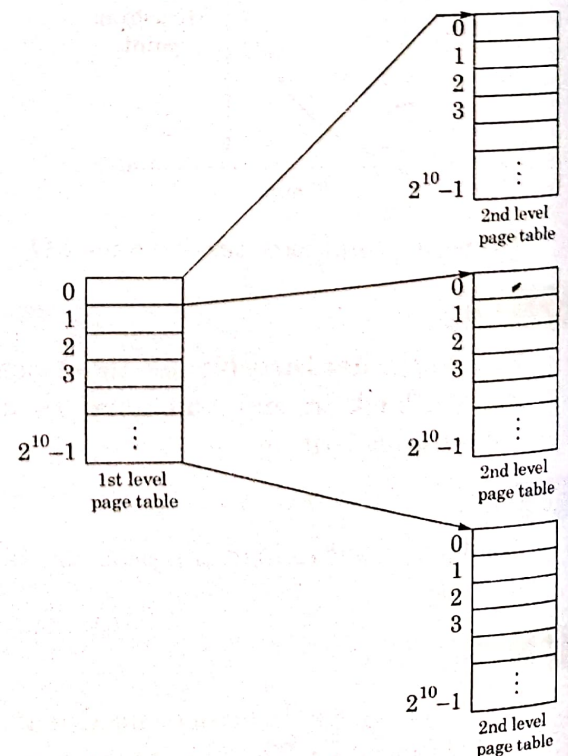


Address will be found in first level page table second block.

Third virtual address given



Address will be found in first level page table last block.



To execute the process we need to bring all the above 4 page tables in main memory.

$$4 \times 2^{10} \times 4 \text{ B} = 16 \text{ kB}$$

4.19 (d)

The I/O performance of a user program determined by many input and output devices not only by the disk. When we use FCFS and replace it by SSTF it improves only disk driver performance not the entire I/O performance. So I/O improvement performance of user program is 0%.

4.20 (a)

Minimum number of page frames allocated to a running process is determined by the instruction set architecture.

4.21 (d)

Effective average instruction execution time = CPU time (100 ns) + Average time lost due to

page fault per instruction $\left(= \frac{8 \text{ ms}}{10000} \right) + \text{Number}$

of memory references $\times \text{EMAT}$

$100 + 800 + 2 \times [0.9 (0 + 150) + 0.1 (0 + 3 \times 150)]$
= 1260

4.22 (b)

Fully associate cache with 8 block i.e. 1 set with 8 cache blocks.

Set	4	3	25	8	19	6	25	16
	B_0	B_1	B_2	B_3	B_4	B_5	B_6	B_7

Since set contain 8 blocks so, request for 45 replacement is needed. When 45 comes then

45	3	25	8	19	6	25	16
B_0	B_1	B_2	B_3	B_4	B_5	B_6	B_7

For request 22:

45	22	25	8	19	6	25	16
B_0	B_1	B_2	B_3	B_4	B_5	B_6	B_7

For request 8. Since already present, so no replacement.

For request 3:

45	22	25	8	3	6	25	16
B_0	B_1	B_2	B_3	B_4	B_5	B_6	B_7

For request 16; since already present, so no replacement.

For request 25; since already present, so no replacement.

For request 7:

45	22	25	8	3	7	25	16
B_0	B_1	B_2	B_3	B_4	B_5	B_6	B_7

So block 5 i.e. B_5 contain memory block 7.

4.23 (d)

In page table, number of entries is equal to number of pages in virtual memory and page table entry is equal to number of frames and other information i.e.

So, number of frames = $\frac{\text{Physical memory}}{\text{Page size}}$

$$= \frac{2^{30}}{2^{12}} = 2^{18}$$

So, number of bits for frames = $\log_2 (2^{18}) = 18$

We know that,

Page table entry = Number of bits for frames + Other information

$$32 = 18 + x$$

$$x = 32 - 18 = 14$$

So, 14 bits can be used for storing protection and other information.

Note: Each page table entry must contain frame number.

4.24 (b)

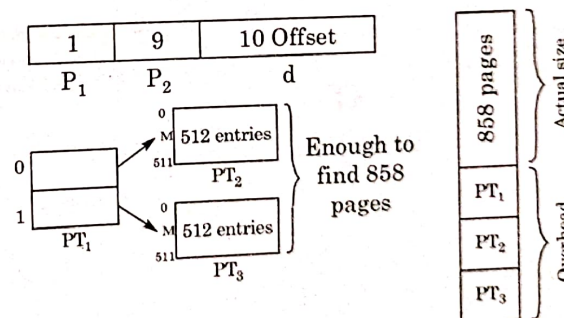
Overhead is considered as memory wasted by maintaining the page tables.

So, we need to find the overhead for every case in the given question.

Case-1: Overhead using two level paging.

Each page size is 1 KB.

Total pages = $195 + 254 + 45 + 364$
= 858 KB pages



Each page table entry = 4 bytes.

Size of page table₁ = $2 \times 4 = 8$ bytes

Size of page table₂ = $2^9 \times 4 = 2048$ bytes

Size of page table₃ = $2^9 \times 4 = 2048$ bytes

$\therefore$ Overhead of paging (P) = $8 + 2048 + 2048$
= 4104 bytes

Case-2: Overhead using segmentation.

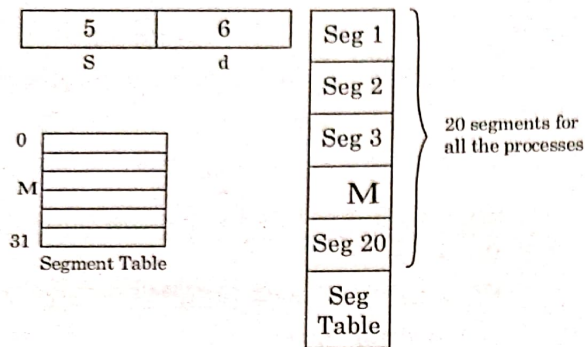
Number of segments = $4 + 5 + 3 + 8 = 20$ (5 bits for segment number)

Maximum segment size

$$= \max \left(\frac{195}{4}, \frac{254}{5}, \frac{45}{3}, \frac{364}{8} \right)$$

$= \max(49, 51, 15, 46) = 51$ bytes (helps to find segment offset)

Number of bits required to find the limit (segment offset) = 6



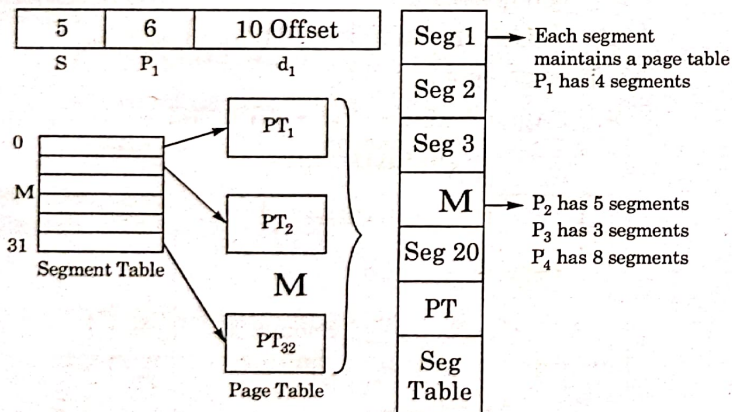
Size of segment table (approx)
 $= 32 \times 8 = 256$ bytes

$\therefore$ Overhead using segmentation (S) = 256 bytes

Case-3: Overhead using segmentation and paging.

Number of segments = 20 $\Rightarrow$ 5 bits required for segment number.

Maximum segment size = 51 bytes $\Rightarrow$ 6 bits required for page offset.



20 page tables are required.

Each segment needed 1 page table.

Size of segment table = $32 \times 8 = 256$ bytes.

Size of each page table = $64 \times 4 = 256$ bytes.

$\therefore$ Segmentation and paging overhead (T)

$$= 256 \text{ B} + 20 \times 256 \text{ B} = 5376 \text{ B}$$

S: Simple segmentation takes less overhead.

P: Two level paging takes little more overhead because of two page tables.

T: Segmented paging takes more overhead because we have paging on both segment and segment table.

So, $S < P < T$.

4.25 (c)

Number of bits for virtual address = 32

Number of bits for pages = 4 KB

$$= \log_2(2^{12}) = 12$$

So, number of bits for pages = $32 - 12 = 20$

$$\text{Number of sets} = \frac{128}{4} = 32$$

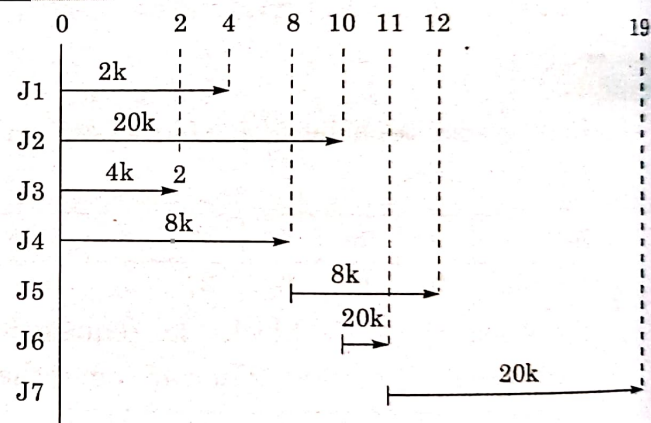
Number of bits for sets = $\log_2(32) = 5$

So, minimum tag size = $20 - 5 = 15$

4.26 (a)

The main purpose of virtual memory is by allocating less memory to the process, we plan to increase degree of multiprogramming, by bringing more processes into memory. If virtual memory is removed then the efficient implementation of multiprogramming and multi user is no longer possible.

4.27 (b)



4.28 (c)

Size of page = 100

When page fault occur then from that page to next 99 byte are fetched into memory

Reference address	Address in memory	Hit/Fault
0100	0100 - 0199	Fault
0200	0200 - 0299	Fault
0430	0430 - 0529	Fault
0499	0430 - 0529	Hit
0510	0430 - 0529	Hit
0530	0530 - 0629	Fault
0560	0530 - 0629	Hit
0120	0120 - 0219	Fault
0220	0220 - 0319	Fault
0240	0220 - 0319	Hit
0260	0220 - 0319	Hit
0320	0320 - 0419	Fault
0410	0320 - 0419	Hit
		Total fault = 7

4.29 (a)

Average Access Time = $(1 - p)(T_{\text{mem}}) + p[p(\text{S.T. dirty}) + (1 - p)(\text{S.T. not dirty})]$

$$3 = (1 - p)(1) + p(p \times 300) + (1 - p) \times 100$$

$$3 = 1 - p + p^2 300 + p 100 - p^2 100$$

$$3 = 1 + 99p + 200p^2$$

$$200p^2 + 99p - 2 = 0$$

$$p = \frac{b \pm \sqrt{b^2 - 4ac}}{2a} = 0.0194$$

4.30 (a)

Number of page frames = 3

Consider the optimal page replacement algorithm as follows: 1, 2, 1, 3, 7, 4, 5, 6, 3, 1

10 Page Frames	*	*	*	3	3	3	3	3	3	3
Referring String	1	2	1	3	7	4	5	6	3	1
Page	Page	Page	Page	Page	Page	Page	Page	Page	Page	Page
	Fault	Fault		Fault	Fault	Fault	Fault	Fault		

So total number of page faults = 7

4.31 (c)

Consider the LRU page replacement policy

10 Page Frames				3	3	3	5	5	5	1
Referring String	1	2	1	3	7	4	5	6	3	1
Page	Page	Page	Page	Page	Page	Page	Page	Page	Page	Page
	Fault	Fault		Fault	Fault	Fault	Fault	Fault	Fault	Fault

So total number of page fault = 9

Number of page fault in LRU – Number of page fault in optional = $9 - 7 = 2$ more page fault.

4.32 (c)

Effective access time = hit ratio * (TLB access time + Memory access time) + miss ratio * (TLB access time + Page table access time + Memory access time)

$$= 0.9 * (10 + 50) + 0.1 * (10 + 50 + 50)$$

$$= 54 + 11 = 65$$

4.33 (b)

Page tables are stored in the main memory and main memory is divided into frames, it means addressing the page table is same as addressing the frames, so we just need to find, the number of bits required for addressing frames.

Number of bits required for first level = 24

Number of bits required for second level = 24

Number of bits required for third level = 24

4.34 (b)

Main memory is divided in page size = 16 bytes

We know that, page size = frame size

So in virtual address least 4 bit represent page size and remaining bits represents page number.

So, page number request are:

0 = 00 0000 i.e. Page number = 0

4 = 00 0100 i.e. Page number = 0

8 = 00 1000 i.e. Page number = 0

20 = 01 0100 i.e. Page number = 1

24 = 01 1000 i.e. Page number = 1

36 = 100 1000 i.e. Page number = 4

44 = 10 1100 i.e. Page number = 2

12 = 00 1100 i.e. Page number = 0

68 = 100 0100 i.e. Page number = 4

72 = 100 1000 i.e. Page number = 4

80 = 101 0000 i.e. Page number = 5

84 = 101 0100 i.e. Page number = 5

28 = 1 1100 i.e. Page number = 1

32 = 10 0000 i.e. Page number = 2

88 = 101 1000 i.e. Page number = 5

92 = 101 1100 i.e. Page number = 5

Request are:

0	0	0	1	1	4	2	0	4	4	5	5	1	2	5	5
						2	2	2	2	2	2	1	1	1	1
					4	4	4	4	4	4	4	4	4	4	4
			1	1	1	1	1	1	1	5	5	5	5	5	5
0	0	0	0	0	0	0	0	0	0	0	0	0	2	2	2
F	H	H	F	H	F	F	H	H	H	F	H	F	F	H	H

So, at the end pages 1, 2, 4, 5 are in memory and 7 page fault are present.

4.35 (d)

- Dirty bits used in write back policy.
- R/W bits used in page protection.
- Reference bit used in page replacement policy (second chance policy).
- Valid bit used when page is initialized.

4.36 (a)

In Bledy's Anomaly if number of frames are increased then number of page faults increases. This behaviour found only with FIFO.

4.37 (b)

Page frame numbers are most important, virtual page number may not be stored entirely.

4.38 (a)

FIFO policy for page replacement used.
Access 100 distinct pages by taking some example: 2 3 4 5 6 7 8 9
So by loading it get

5	9
4	8
3	7
2	6

= 4 page fault, = 4 page fault

and now access these page in reverse so

9 8 7 6 5 4 3 2
So no page fault for these page fault for these

9	5
8	4
7	3
6	2

4 page fault

So total = 4 + 4 + 4 = 12 page fault

For 8 pages = $2 \times 8 - 4 = 12$

So for n pages = $2n - 4$

So for 100 pages = $2 \times (100) - 4 = 196$

4.39 (b)

Effective access time = $[(1-p) \times \text{access time when no page fault} + p \times \text{access time during page fault}]$

$$= \left[1 - \left(\frac{1}{10^6} \right) \right] \times 20 \text{ ns} + \left(\frac{1}{10^6} \right) \times 10 \text{ ms}$$

$$= \left(\frac{10^6 - 1}{10^6} \right) \times 20 \text{ ns} + \left(\frac{1}{10^6} \right) \times 10 \times 10^6 \text{ ns} \approx 30 \text{ ns}$$

4.40 (b)

OPTIMAL										FIFO									
		3	3	3	3	2	2	2			3	3	3	3	3	2	2	2	
	2	2	2	4	4	4	4	4		2	2	2	2	1	1	1	1	1	
1	1	1	1	1	1	1	1	1	1	1	1	1	1	4	4	4	4	4	4
MISS	MISS	MISS	MISS	MISS	HIT	H	M	H	H	M	M	M	H	M	M	H	M	H	H
5 miss										6 miss									

LRU

		3	3	3	1	1	1	4	4
	2	2	2	2	2	3	3	3	1
1	1	1	1	4	4	4	2	2	2
M	M	M	H	M	M	M	M	M	M

⇒ 9 MISS

So optimal < FIFO < LRU

4.41 (c)

Assume Page Size = x bytes

We know that,

1st Page table size = Number of entries in page table × Page table entry size

$$= \frac{2^{46}}{x} \times 4B = \frac{2^{48}}{x}$$

2nd Page table size = Number of entries in 1st page table × Page table entry size

$$= \frac{2^{48}}{x^2} \times 4B = 2^{50}$$

3rd Page table size = Number of entries in 2nd page table × Page table entry size

$$= \frac{2^{50}}{x^3} \times 4B = \frac{2^{52}}{x^3}$$

Since 3rd level page table must fit into single page.

$$\text{So, } x = \frac{2^{52}}{x^3}$$

$$x^4 = 2^{52}$$

Take log both side

$$4(\log_2 x) = 52 (\log_2 2)$$

$$(\log_2 x) = 13$$

$$x = 2^{13} = 8 \text{ KB}$$

4.42 (c)

Since each cache line can contain 16,648 blocks, assuming best case 1024 bytes can accommodate

$$\text{in one line hence } \frac{2^{13}}{2^{10}} = 2^3$$

⇒ Number of colours needed = 8.

4.43 (7)

Optimal page replacement policy:

✓	✓	✓	✓	*	*	✓	✓	*	*	✓
		3	4	4	4	4	4	4	4	6
	2	2	2	2	2	2	2	2	2	2
1	1	1	1	1	1	5	3	3	3	3
1	2	3	4	2	1	5	3	2	4	6

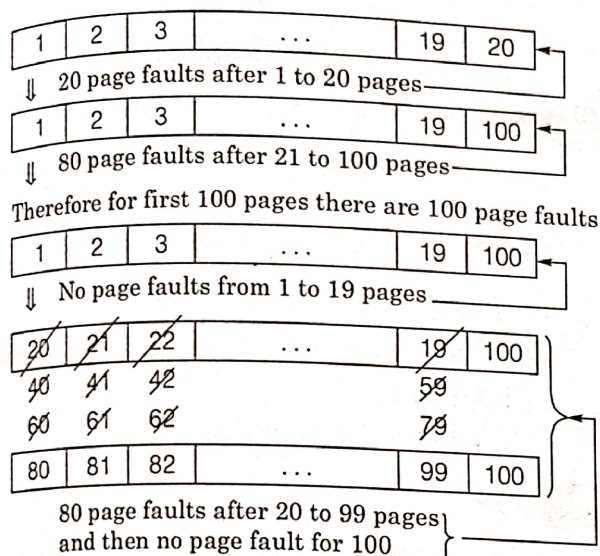
∴ 7 page faults

4.44 (d)

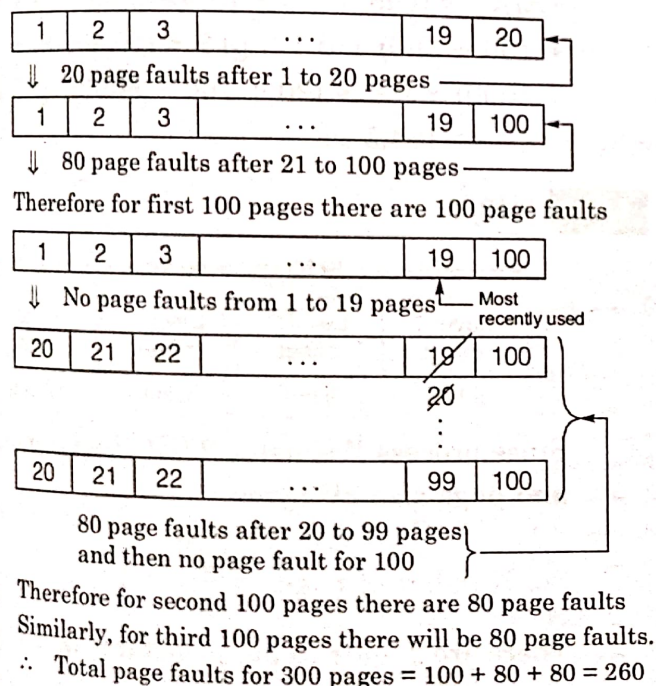
Pages:

1, 2, ..., 20, ..., 100, 1, ..., 20, ..., 100, 1, ..., 20, ..., 100,

Optimal page replacement:



Most recently used:



4.45 (6)

✓	✓	✓	✓	✗	✗	✗	✓	✓	✗
		6	6	6	6	6	6	7	7
		7	7	7	7	7	2	2	2
4	4	4	1	1	1	1	1	1	1
4	7	6	1	7	6	1	2	7	2

∴ 6 page faults will occur using LRU

4.46 (122)

$$\text{Effective Memory Access Time} = H_{\text{TLB}} (T_{\text{TLB}} + T_M) + (1 - H_{\text{TLB}}) (T_{\text{TLB}} + T_M + T_M)$$

$$= 0.6 (10 + 80) + (1 - 0.6) (10 + 80 + 80)$$

$$= 0.6 (90) + 0.4 (170) = 122$$

Note: In case of TLB miss 1st T_M is page table access time while 2nd T_M is main memory access time.

4.47 (4)

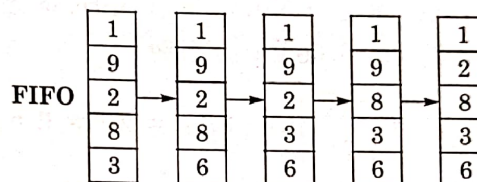
Page table size = Number of page table entries × Entry size

$$= \text{Number of pages} \times 4 \text{ bytes}$$

$$= \frac{2^{32}}{2^{12}} \times 4 \text{ bytes} = 4 \text{ MB}$$

4.48 (a)

3, 8, 2, 3, 9, 1, 6, 3, 8, 9, 3, 6, 2, 1, 3



FIFO 9 page faults.

LRU														
				8	2	2	9	1	1	1	8	9	3	
			8	2	3	9	1	6	6	8	9	3	9	
		3	2	3	9	1	6	3	8	9	3	6	6	
	3	8	3	9	1	6	3	8	9	3	6	2	2	
3	8	2	9	1	6	3	8	9	3	6	2	1	1	

Here LRU also incurs 9 page faults.

Both FCFS and LRU incurs the same number of page faults.

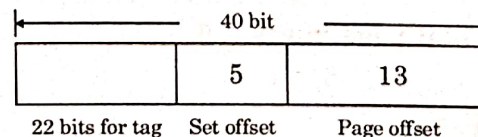
4.49 (22)

Page size = 8 kB = 2^{13} bytes

⇒ 13 bits required for offset.

Number of sets in cache = 32

⇒ 5 bits required for index (set offset).



∴ 22 tag bits are required.

4.50 (a)

The best fit algorithm finds the smallest sufficient partition.

	357	491	468		210
200 KB	400 KB	600 KB	500 KB	300 KB	250 KB

∴ The memory partitions 200 KB and 300 KB are not allotted.

4.51 (36)

Page size = 8 KB $\Rightarrow$ 13 bit offset
 Number of frame bits = 32 - 13 = 19 bits
 Page table entry = Valid + Dirty + Permission
 bits + Translation (frame bits)
 $= 1 + 1 + 3 + \text{Frame bits}$
 $= 5 + 19 = 24 \text{ bits}$

Page table size = 24 Mbytes

Number of pages = Number of page table entries

$$= \frac{24 \text{ Mbytes}}{24 \text{ Bits}} = 8 \text{ M} = 2^{23} \text{ pages}$$

$\therefore$ 23 bits needed for page and 13 bits offset

Length of virtual address = 23 + 13 = 36 bits.

4.52 (384)

Page table size = Number of entries in page
 table $\times$ Page table entry size

$$= \left(\frac{2^{40}}{2^{14}} \right) \times 48 \text{ bits} = 2^{26} \times 6 \text{ bytes}$$

$$= 64 \text{ M} \times 6 \text{ B} = 384 \text{ MB}$$

4.53 (1)

This question is little tricks, but not difficult.
 You can solve the problem by taking smaller
 example with smaller values.

Example: 1, 2, 3, 4, 1, 2, 3, 4 with '2' frames.

LIFO: 1, 2, 3, 4, 1, 2, 3, 4

	1	2	3	4	1	3	4
1	1	1	1	1	1	1	1
F	F	F	F	F	F	F	F

Total page faults = 7

Optimal: 1, 2, 3, 4, 1, 2, 3, 4

	1	2	4	4	4	4	4
1	1	1	1	1	2	3	3
F	F	F	F	F	F	F	F

Total page faults = 6

$$7 - 6 = 1$$

4.54 (d)

Because of Belady's anomaly, it will happen in
 FIFO phase replacement algorithm.

4.55 (b)

Considering each statement:

- Random page replacement algorithm can behave like any algorithm probably FCFS too, hence it can suffer from Belady's anomaly.
- LRU page replacement algorithm doesn't suffer from Belady's anomaly.

4.56 (b)

$$\text{EMAT} = P \times S + (1 - P) \times M$$

$$X = P \times D + (1 - P) \times M$$

$$X = P \times D + M - P \times M$$

$$X - M = P(D - M)$$

$$P = \frac{X - M}{D - M}$$

4.57 (d)

1 word = 4 bytes, Page size = 8 kB = 2^{13} B

$$\text{Number of words in 1 page} = \frac{2^{13}}{2^2} = 2^{11}$$

TLB can hold 128 valid entries so, at most
 128×2^{11} memory address can be addressed
 without TLB miss.

$$128 \times 2^{11} = 256 \times 2^{10}$$

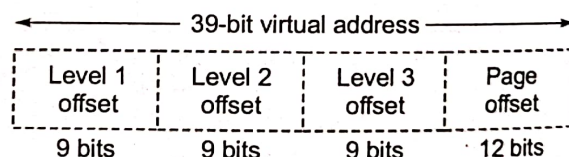
4.58 (c)

The hole created by best fit is never larger than
 the hole created by first fit. This is correct option.

4.59 (154.5)

$$\text{EMAT} = 0.95 \times (20 + 100) + 0.05 \times (0.9 \times (20 + 100 + 100) + 0.1 \times [0.2 \times (20 + 100 + 5000 + 5000) + 0.8 \times (20 + 100 + 5000)])$$

$$= 154.5 \text{ ns}$$

4.61 (4108)

Since process P is using 2 GB virtual memory
 and page size 2^{12} bytes.

$$\text{So, Number of pages} = \frac{2^{31}}{2^{12}} = 2^{19}$$

There are 2^9 entries per page table in 3rd level.

$$\text{So, we need} = \frac{2^{19}}{2^9} = 2^{10} \text{ page tables in 3rd level}$$

Now, it means 2^{10} entries in 2nd level so number
 of page tables in 2nd level

$$= \frac{2^{10}}{2^9} = 2 \text{ page table}$$

Now we have 2 entries in 1st level hence we need
 only 1 page table in 1st level.

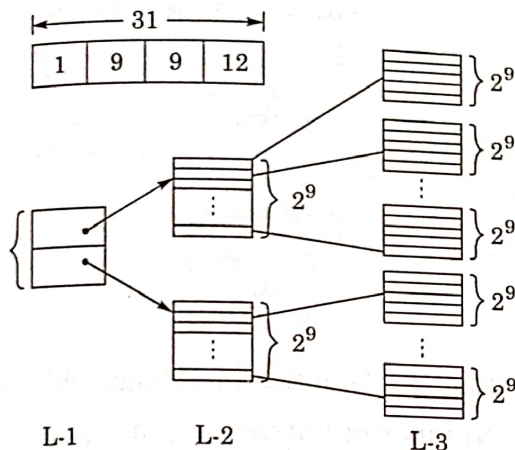
$$\text{Total number of page table} = 2^{10} + 2 + 1$$

$$= 1027 \text{ per table}$$

Each page table has 2^9 entries and each entry size is 8 bytes.

So, total size of page table in bytes
 $= 1027 \times 2^9 \times 8 \text{ bytes} = 4108 \text{ KB}$

Alternative method:



Total page table = $1 + 2^1 + 2^1 \times 2^9 = 1027$

Page table size = $1027 \times 2^{12} = 4108 \text{ KB}$

4.62 (c)

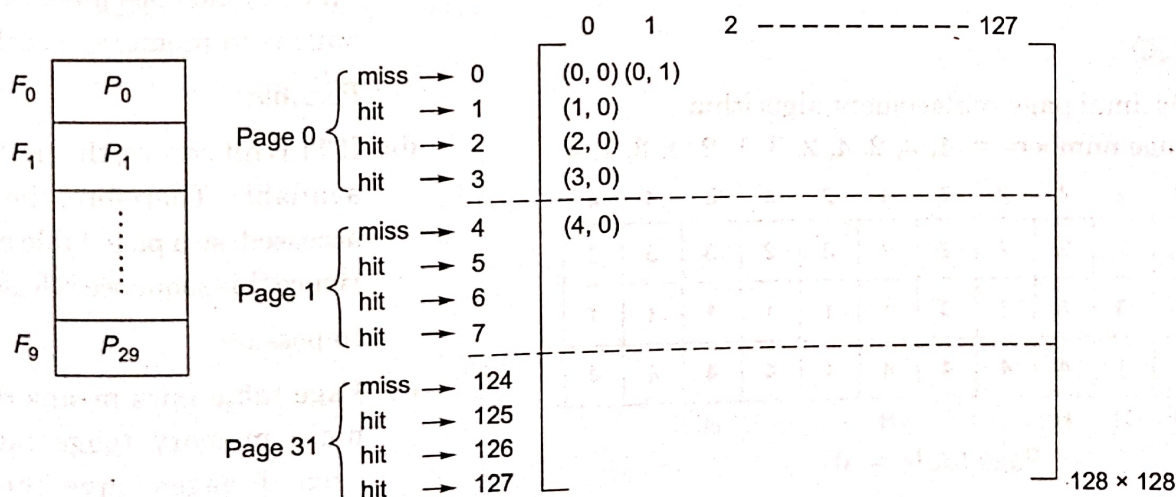
- (a) TLB performs parallel search. Hence true
- (b) If the TLB hit means, word will surely present in main memory. Hence true
- (c) Memory access time using inverted page table is always not same, because there is no indexing applied, we follow no equal linear search. Hence false
- (d) If two distinct virtual address map to same value while hashing, they will be resolved using linked list, so memory access time will not be same. Hence true

4.63 (0.6)

9 page faults out of total 15 references $\frac{9}{15} = 0.6$

4.64 (4096)

Given array dimension is $D[128][128]$ which is stored in row major order. It means that, 128 rows and 128 columns.



Given that, each physical frame holds 512 elements of the array D . 30 page frames are allocated to a process. Array access is in column major order $[D[j]][i] \text{ } i \neq 10$

As each physical frame holds 512 elements, the 1st 4 rows of the array belongs one page, we can see that, one page fault 4 elements.

Array contains 32 pages, but memory contains 30 page frames.

$$\therefore n_{\text{out}} \text{ page faults} = \frac{128 \times 128}{4} = \frac{2^{14}}{2^2} = 4096$$

4.65 (5)

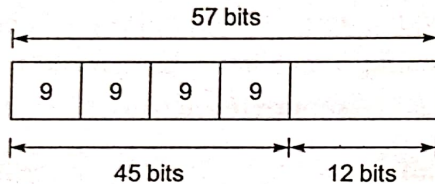
$$\begin{aligned}\text{Offset} &= \log_2(\text{Page size}) \\ &= \log_2(4 \text{ KB}) = 12 \text{ bits} \\ \text{PTE} &= 8 \text{ bytes}\end{aligned}$$

For every level page table should fit into a page.

$$\text{So, number of entries in a page} = \frac{4 \text{ KB}}{8 \text{ B}} = 2^9$$

$$\text{Bits for a level} = \log_2(2^9) = 9 \text{ bits}$$

Following is representation



So, 5 levels required.

4.66 (6596)

$$\text{P.A.} \rightarrow (1100111000100)_2 \rightarrow (6596)_{10}$$

4.69 (a)

$$\text{L.A.} = 32 \text{ bits}$$

$$\text{P.A.} = 20 \text{ bits}$$

$$\text{Page size} = 2048 \text{ byte} = 2^{11} \text{ B}$$

Number of entries in page table

$$= \frac{\text{L.A.S}}{\text{Page size}} = \frac{2^{32} \text{ B}}{2^{11} \text{ B}} = 2^{21}$$

4.70 (6)

Optimal page replacement algorithm

Page numbers = 1, 3, 2, 4, 2, 3, 1, 2, 4, 3, 1, 4

1	3	2	4	2	3	1	2	4	3	1	4
		2	2	2	2	2	2	2	3	3	3
	3	3	3	3	3	1	1	1	1	1	1
1	1	1	4	4	4	4	4	4	4	4	4
H	H	H	H			H			H		

Page fault = 6

4.71 (d)

Reference string 1, 2, 3, 4, 5, 4, 1, 6, 4, 5, 1, 3, 2

Using optimal page replacement policy

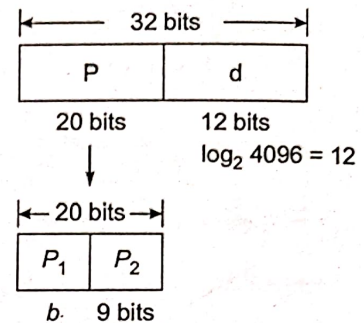
1	2	3	4	5	4	1	6	4	5	3	3	2
		3	3	5	5	5	5	5	5	5	3	3
	2	2	4	4	4	4	4	4	1	1	1	-1
1	1	1	1	1	1	6	6	6	6	6	6	2
H	H	H	H	H		H					H	H

4.72 (11)

$$\text{L.A.S.} = 2^{32} \text{ bits}$$

$$\text{Page size} = 4096 \text{ bytes}$$

Two level paging



Each table in the inner page table takes 8 bytes

Number of entries in inner page table

$$= \frac{4096}{8} = \frac{2^{12}}{2^3} = 2^9$$

$$\text{Out page table index} = 20 - 9 = 11 \text{ bits.}$$

4.73 (b, c)

(a) TLB miss means the translation is not in TLB, so the page table is accessed. Page table hit means the page is present in main memory.

Since cache uses physical address and the page is in memory, a cache hit is possible.

Possible

(b) If TLB hit occurs, the translation is already available. Therefore, the page table is not accessed, so a page table miss cannot occur. Hence this sequence is logically inconsistent.

Impossible

(c) Page table miss means the page is not in main memory (page fault). Given that evicted pages have their cache blocks invalidated, cache cannot contain valid data. Therefore, cache hit cannot occur.

Impossible

(d) TLB miss leads to page table access. Page table miss indicates page fault, so data is not in memory. Since cache blocks are invalidated for evicted pages, cache miss is expected.

Possible

4.74 (1536)

TLB reach = Number of TLB entries $\times$ Page size

$$\text{Number of TLB entries} = \frac{\text{TLB reach}}{\text{Page size}} = \frac{1 \text{ MB}}{4 \text{ KB}}$$

$$= \frac{1 \times 2^{20} \text{ B}}{4 \times 2^{10} \text{ B}} = \frac{2^{20}}{2^{12}} = 2^8 = 256$$

$$\text{Number of pages} = \frac{\text{Virtual address space}}{\text{Page size}}$$

$$= \frac{64 \text{ GB}}{4 \text{ KB}}$$

$$= \frac{2^6 \times 2^{30} \text{ B}}{2^2 \times 2^{10} \text{ B}} = \frac{2^{36}}{2^{12}} = 2^{24}$$

$$\text{Page bits} = \log_2 2^{24} = 24 \text{ bits}$$

$$\text{Number of frames} = \frac{\text{Physical address space}}{\text{Frame size}}$$

$$= \frac{2^{30} \text{ B}}{2^{12} \text{ B}} = 2^{18}$$

$$\text{Frames bits} = \log_2 2^{18} = 18 \text{ bits}$$

TLB entry bits = Process ID + Page number
+ Frame number + Control bit

$$= 4 + 24 + 18 + 2 = 48 \text{ bits}$$

TLB size = TLB entry bits $\times$ bits per entry

$$= \frac{256 \times 48}{8} = 1536 \text{ Bytes}$$

4.75 (3)

1	20
2	4
3	25
4	18 16
5	7
6	9
7	15
8	12

P1 = 16 KB

P2 = 9 KB

2 KB

9 KB

For P1 = 16 KB

By using Best fit

holes $\geq$ 16 KB

Smallest suitable hole = 18 KB (hole 4)

For P2 = 9 KB

By using best fit

holes $\geq$ 9 KB

Smallest suitable hole = 9 KB (hole 6)

Less than 8 KB,

4 KB, 2 KB, 7 KB = 3

■■■■■